

Oct 7 — 2 papers

Episode 21 · October 7, 2026 · 2 papers

Priority Coordination Games: Hodge Decomposition and a Sharp Design Limit

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Zhihao Lin et al.

If you model announced right-of-way priorities at an unsignalised intersection as a potential game, you are guaranteed to throw away a fixed share of the incentive structure, for example one fifth at an eight-vehicle intersection, and no rank-based payoff redesign can recover it.

PROBLEM Analyses of decentralised priority negotiation routinely replace the real game with a potential game, meaning a single shared objective, because it makes convergence and design tractable. Nobody had measured exactly what that substitution loses, or whether payoff design could close the gap.

RESULTS There are no experiments. The results are theorems. The best common-objective fit to the choice log-odds has a relative squared error of at least one over the maximum per-agent conflict count plus one. That bound is exactly one fifth for an eight-vehicle intersection, whatever the number of priority levels. On a complete conflict graph of N agents with at least three levels and a total-order protocol, every nonconstant rank-based payoff leaves at least one over N error, and only affine payoffs reach that floor.

METHOD They split the agents' incentives with the Hodge decomposition into a potential part, which a common objective can express, and a harmonic, cyclic part, which it cannot. For a linear payoff they derive both parts in closed form on any conflict graph and under any deterministic tie-breaking rule. The harmonic energy equals the number of conflicts. The potential energy adds the number of adjacent conflict pairs. They then connect the harmonic part to the stationary probability current of low-rationality log-linear learning, and prove a lower bound that holds for any rank-based payoff on the complete conflict graph.

CAVEAT The headline error is measured on choice log-odds weighted uniformly over unilateral moves. Nothing shows that this error translates into worse throughput, safety or deadlock rates in a real merge or intersection. The dynamic interpretation holds only to leading order in the low-rationality limit, with uniform revision. The paper also notes that the missed component is invisible to strict-improvement dynamics, which is the kind of better-response negotiation most vehicle protocols actually run. The model abstracts away vehicle dynamics, timing and message loss entirely. Priority is a one-shot level announcement served in order, so there is no proposal-acceptance exchange and no partial participation.

WHY IT MATTERS This bears directly on the conflict-resolution and game-theoretic right-of-way part of your work. If you justify a negotiation scheme by assuming a potential game, this paper gives a citable, quantitative limit on that assumption, and it tells you that reshaping payoffs will not fix it. It says nothing about message design, platooning, trajectory-level coordination or communication degradation. Its practical value depends on whether your agents use noisy learning rules, where the cyclic component matters, or strict best response, where it does not.

arXiv:2610.05832v1 PDF [notable](#)

Cooperative Adaptive Cruise Control with Constant Distance Gaps for Retrofitted Legacy Vehicles

SKIM

Kay Massow et al.

A team retrofitted ordinary production cars with MPC-based cooperative adaptive cruise control and drove them off from a traffic light with gaps under three meters, with each car listening only to the car directly ahead.

PROBLEM Theory says a constant-distance spacing policy that uses only the predecessor's data cannot be string stable, so close standstill gaps that would raise intersection discharge rates were thought to be infeasible on real vehicles. The intersection benefit also needs high penetration, which means the system has to work on existing combustion cars with heterogeneous, proprietary powertrains.

RESULTS The abstract claims gaps below three meters in real experiments on heterogeneous retrofitted cars and calls this a first demonstration, but it gives no platoon lengths, no error or overshoot figures, and no comparison between the two relaxation strategies. The intersection capacity gain comes from the authors' earlier simulation study and is not measured here.

WHY IT MATTERS This touches your platooning and string-stability interests, but only from the control side. It deliberately avoids explicit coordination, negotiation, joining, and splitting, so it adds nothing to maneuver protocols or message design. It is worth a look only if you need real-vehicle evidence on how tight a gap predecessor-only communication can hold, or a cheap baseline to compare against.

METHOD Each car runs a model predictive spacing controller built on a first-order longitudinal model. It is paired with a powertrain feedforward that uses each manufacturer's own torque model, so no confidential vehicle parameters are shared. Communication is one-vehicle look-ahead only, with no platoon management layer. To stop disturbances from growing along the string, they compare two ways of loosening the gap a little: a tiny time gap near the string-stability limit, and a gap that stretches with acceleration, modeled as a virtual spring and damper. They also define a new metric, desired acceleration overshoot, which links how much each car amplifies actuator demand to how long the platoon can get.

CAVEAT Platoon length is explicitly limited, and the abstract never says how many vehicles were driven, so the headline gap may only hold for a few cars in low-speed startup from a stop. The new overshoot metric is the authors' own yardstick, which makes it hard to compare against standard string-stability measures. Nothing in the abstract tests packet loss, latency, or mixed traffic, which is exactly where a predecessor-only constant-gap scheme is most fragile.

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